UNIVERSITY OF SOUTH CAROLINA

ITEC 493: IT Security for Managers

Introduction to Cybersecurity

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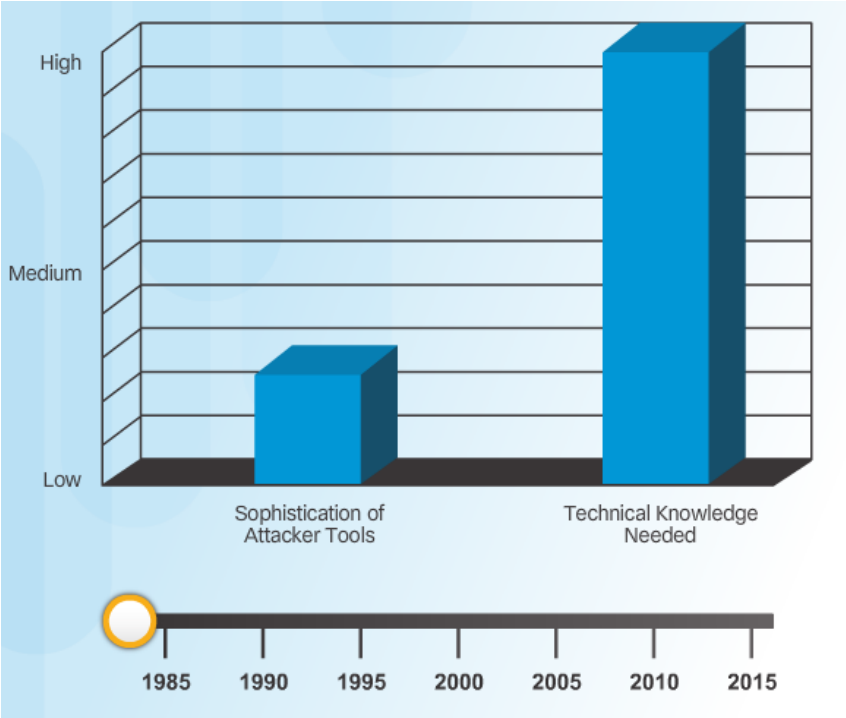
Objectives

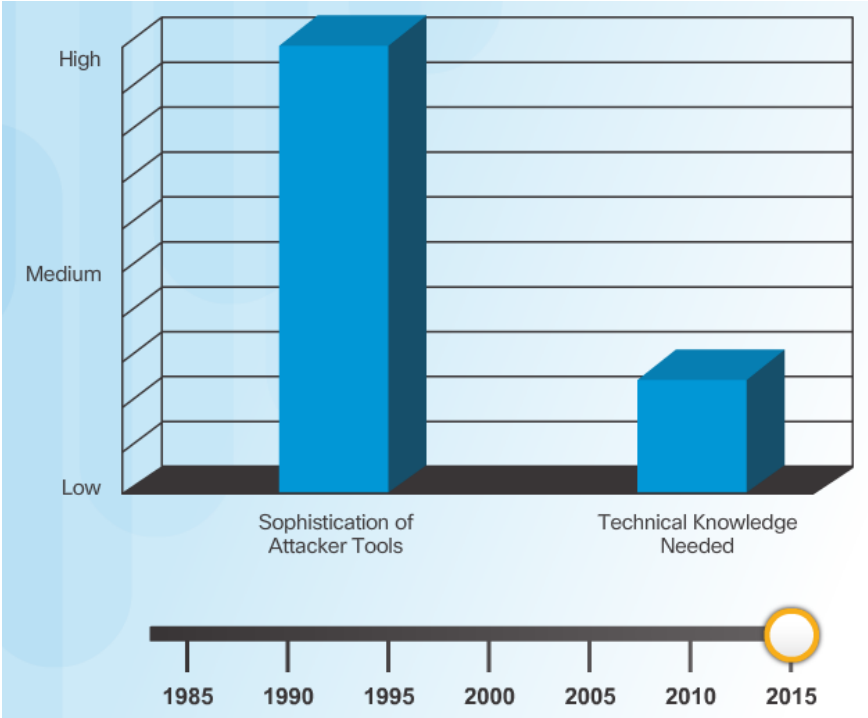
- Describe the challenges of securing information
- Define information security and explain why it is important
- Identify the types of attackers that are common today
- List the basic steps of an attack
- Describe the five basic principles of defense

Difficulties in Defending

Reason	Description
Universally connected devices	Attackers from anywhere in the world can send attacks.
Increased speed of attacks	Attackers can launch attacks against millions of computers within minutes.
Greater sophistication of attacks	Attack tools vary their behavior so the same attack appears differently each time.
Availability and simplicity of attack tools	Attacks are no longer limited to highly skilled attackers.
Faster detection of vulnerabilities	Attackers can discover security holes in hardware or software more quickly.
Delays security updating	Vendors are overwhelmed trying to keep pace updating their products against the latest attacks.
Weak security update distribution	Many software products lack a means to distribute security updates in a timely fashion.
Distributed attacks	Attackers use thousands of computers in an attack against a single computer or network.
Introduction of BYOD	Organizations are having difficulty providing security for a wide array of personal devices.
User confusion	Users are required to make difficult security decisions with little or no instruction.

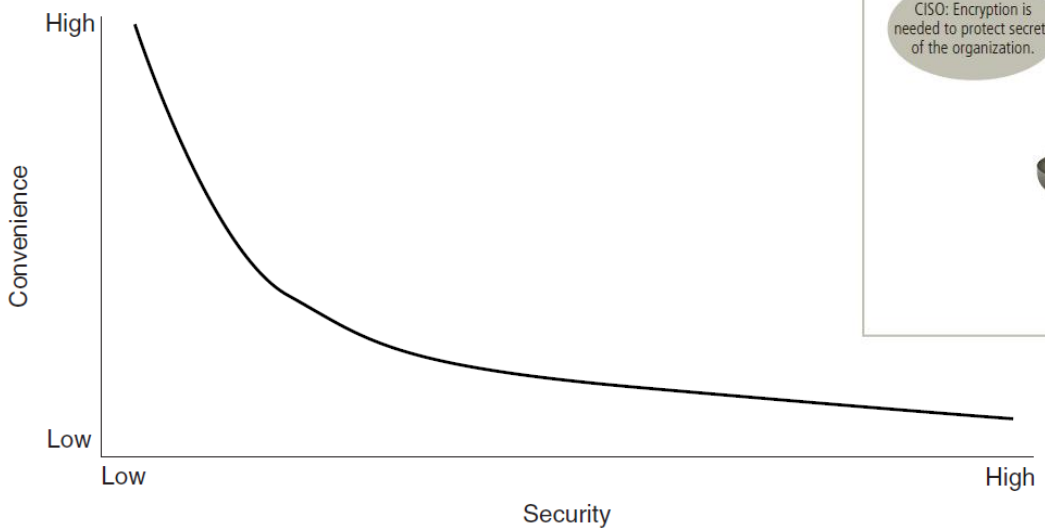
Difficulties in Defending

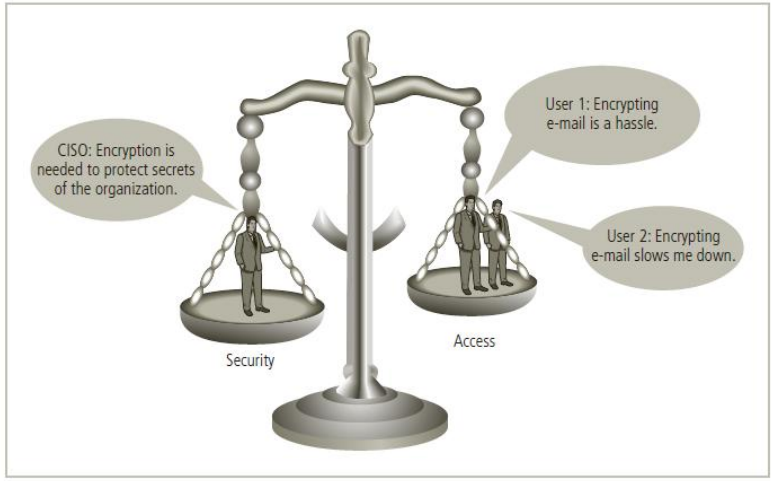




Understanding Security

- Security can be defined as a *process* (how to achieve security) or as a *goal* (what it means to have security)
- Maybe both? The goal to be free from danger as well as the process to achieve that freedom
- Security is the necessary steps to protect a person or property from harm





Goal of Information Security

- Ensure that protective measures are implemented to ward off attacks and prevent the system collapse when a successful attack occurs
- Information security cannot completely guarantee that a system is totally secure

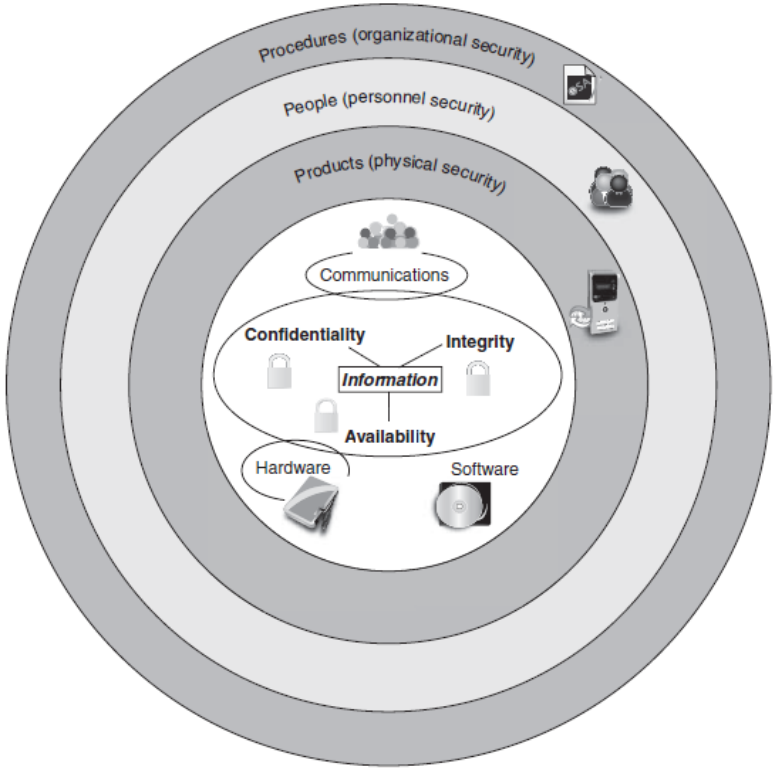
Data Protection

- Data is likely to be an organization's most valuable asset
- How can we protect it?
  - **Ensuring confidentiality**: only authorized parties can access information
  - **Ensuring integrity**: information is not altered
  - **Ensuring availability**: information is accessible when needed
- Also, AAA must be employed
  - **Authentication**: the individual is who he/she claims to be
  - **Authorization**: Providing permission to specific resources
  - **Accounting**: Provides tracking of events

Security Layers

Data stored (at rest) by hardware, manipulated by software, and transmitted by communications, must be protected

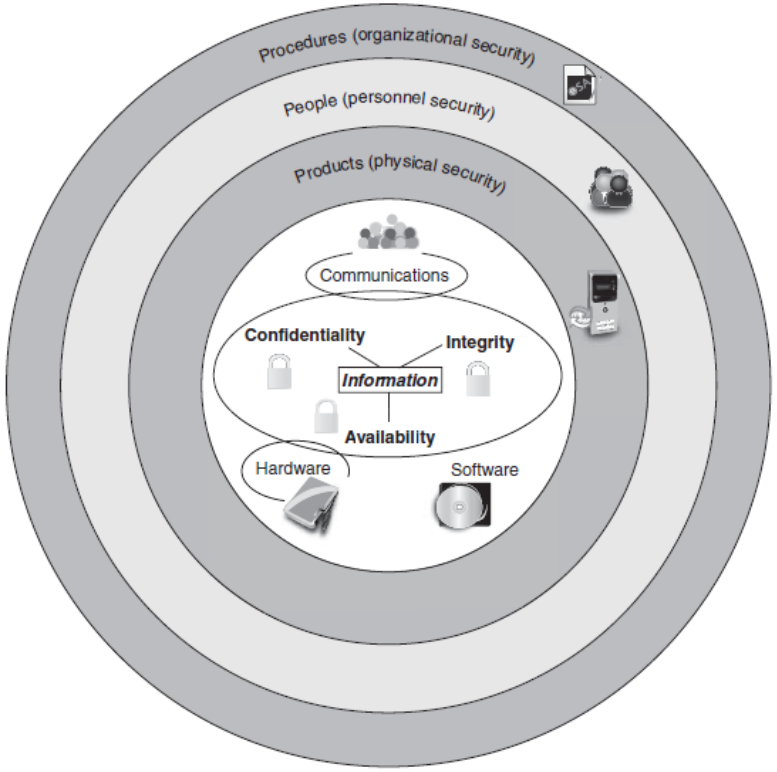
- Policies and procedures
  - Written instructions for accomplishing a specific task
  - Plans and policies in place to ensure people correctly use the products
- People
  - Those who implement and properly use security products to protect data
- Products
  - Security around the data (door locks, firewalls, intrusion prevention system)
  - Security challenges emerge the information systems is connected to networks / Internet



Information Security Definition

- Information security defined as

  "that which protects the confidentiality, integrity, and availability of information on the devices that store, manipulate, and transmit the information through products, people, and procedures"



Technology Assets

- An item that has value
  - Provide value to the organization
  - Cannot easily be replaced without a significant investment in expense, time, worker skill, and/or resources

Element name	Description	Example
Information	Data that has been collected, classified, organized, and stored in various forms	Customer, personnel, production, sales, marketing, and finance databases
Customized business software	Software that supports the business processes of the organization	Customized order transaction application
System software	Software that provides the foundation for application software	Operating system
Physical items	Computers equipment, communications equipment, storage media, furniture, and fixtures	Servers, routers, DVDs, and power supplies
Services	Outsourced computing services	Voice and data communications

Technology Assets

- An item that has value
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Element name	Description	Example	Critical asset?
Information	Data that has been collected, classified, organized, and stored in various forms	Customer, personnel, production, sales, marketing, and finance databases	Yes: Extremely difficult to replace
Customized business software	Software that supports the business processes of the organization	Customized order transaction application	Yes: Unique and customized for the organization
System software	Software that provides the foundation for application software	Operating system	No: Can be easily replaced
Physical items	Computers equipment, communications equipment, storage media, furniture, and fixtures	Servers, routers, DVDs, and power supplies	No: Can be easily replaced
Services	Outsourced computing services	Voice and data communications	No: Can be easily replaced

Threats

- Information security threats

  Events representing a danger to information assets

- The potential for creating a loss is real

  Corruption or theft of information, a delay in information being transmitted, loss of reputation, etc.

- Threat agent

  Person or element (e.g., malicious software) that has the power to carry out a threat

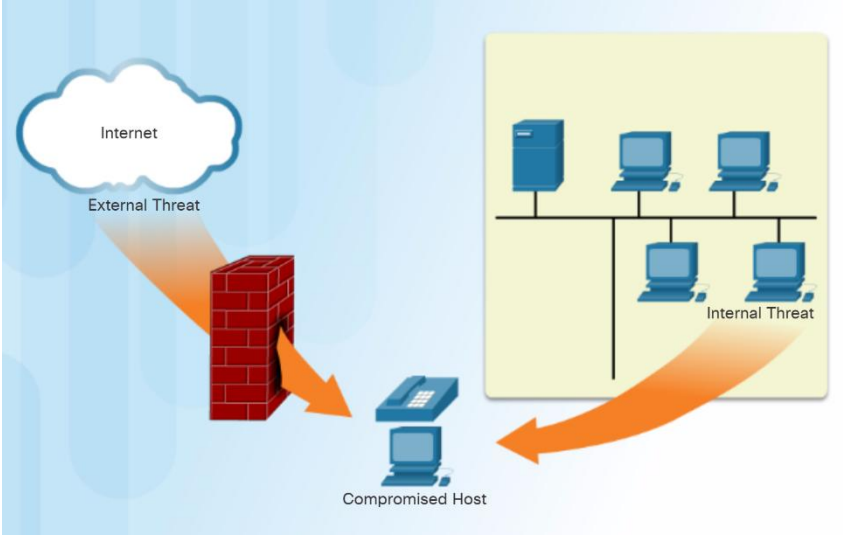
Information Security Terminology: Vulnerability

- **Vulnerability** is a weakness that allows a threat agent to bypass security

  A software defect that allows an unauthorized user to gain control of a computer without the user's knowledge or permission

Information Security Terminology: Threat Vector

- A threat vector is a path or other means by which an attacker can gain access to a server, host, or network
- An attacker, knowing that a flaw in a web server's OS has not been patched, is using the threat vector (exploiting the vulnerability) to steal user passwords
- Examples?



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- Web - fake sites;
- Email – link, attachments
- Removable media
- Almost 94 percent of all malware is delivered through email to an unsuspecting user

Who Are the Attackers?

- Hacker – skilled

  Older term for someone who uses advanced knowledge to attack computers
- Black hat hackers – personal gain

  Attackers who violated computer security for personal gain or to inflict malicious damage
- White hat hackers – received permission

  “Ethical attackers” who received permission to probe system for any weaknesses
- Gray hat hackers – no permission, disclose vulnerability

  Attackers who would break into a computer system without permission and then publically disclose vulnerability

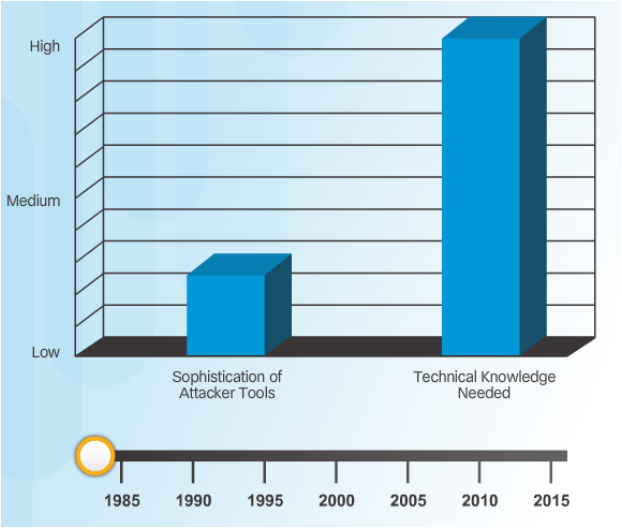
Who Are the Attackers?

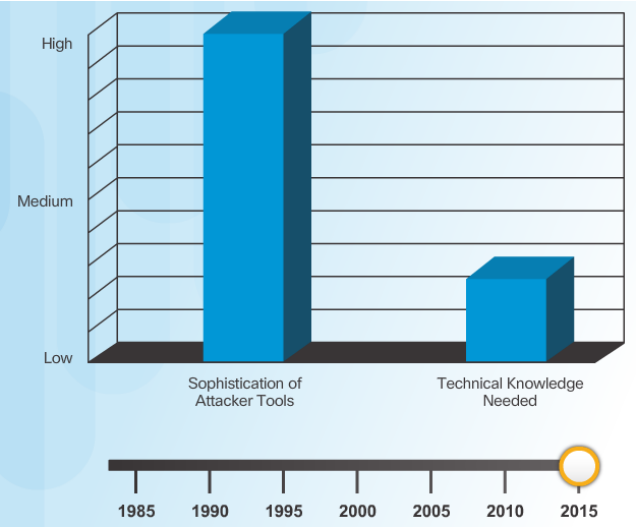
- Cybercriminals
  - Launch attacks against other users and their computers
  - == attackers
  - Highly motivated, less risk-averse, well-funded
  - Goal is financial gain; steal information to generate income
  - Can launch advanced persistent threats (APTs)

    APT ~ unauthorized access to a network and stays there undetected for a long period of time
- More about APT
  - https://www.paloaltonetworks.com/features/apt-prevention

Who Are the Attackers?

- Script kiddies
  - Unskilled users; goal: breaking into computers (damage)
  - Download automated hacking software (scripts)
  - Attack software today has attack capabilities that are even easier for unskilled users; ~40% of attacks performed by script kiddies





Who Are the Attackers?

- Brokers
  - Uncover vulnerabilities and do not report them to vendor
  - Sell vulnerability to other attackers, governments
  - Buyers pay a high price because this vulnerability is unknown

Who Are the Attackers?

- Insiders
  - Employees, contractors, partners who steal from employer
  - Sabotage or theft of intellectual property
  - Employees believing that accumulated data is owned by them

Who Are the Attackers?

- Cyberterrorists
  - Ideologically motivated
  - Perform attacks because of their beliefs
  - Target example: computers that control the electrical power grid of a state or region

Who Are the Attackers?

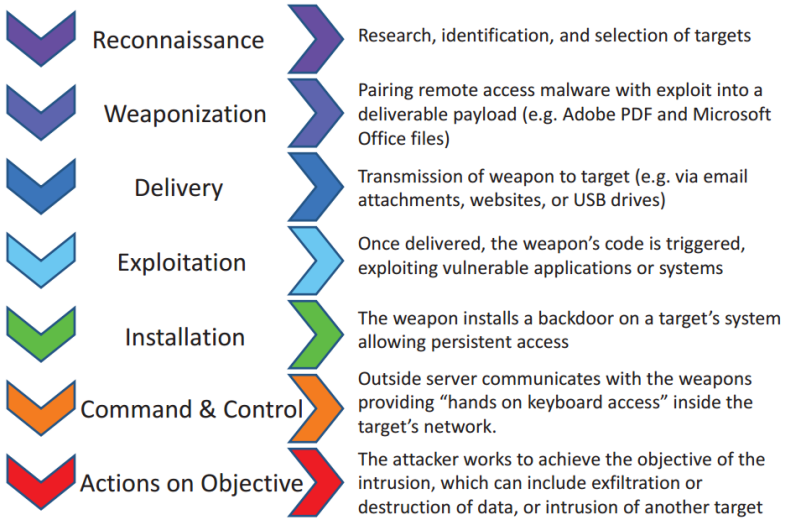
- Hactivists
  - Ideologically motivated
  - Unlike cyberterrorists who incite panic, hactivists are generally not well defined
  - Attacks can involve breaking into a website and changing the contents on the site to make a political statement
  - Retaliatory attacks
  - E.g., Anonymous, Wikileaks
  - Who (if any) should be punished for disclosing information?

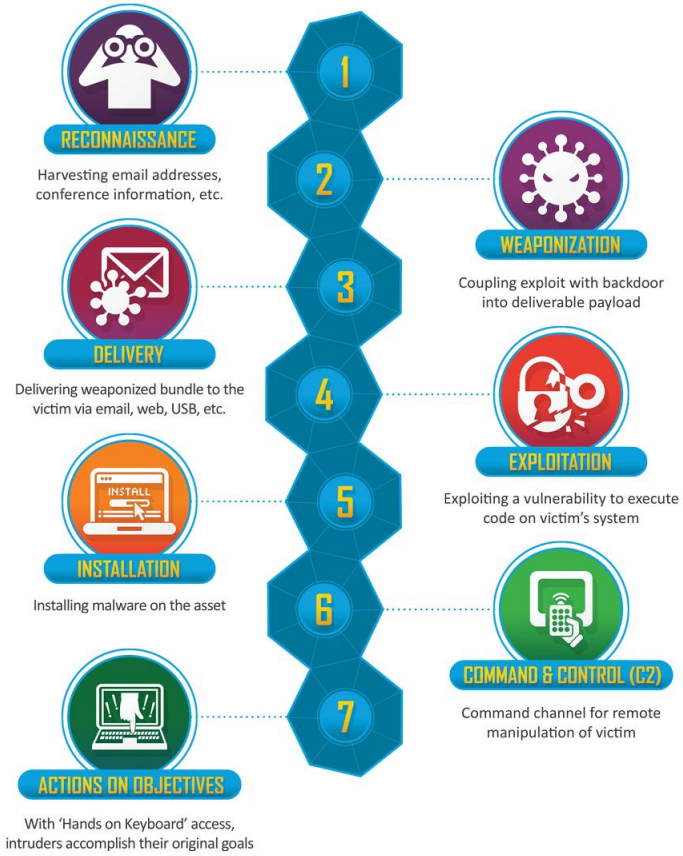
Who Are the Attackers?

- State-sponsored attackers
  - Supported by governments
  - Attackers target foreign governments or even citizens of the government who are considered hostile or threatening
  - Do countries participate (to some degree) in state-sponsored hacking?

Steps of an Attack

- Kill chain





An exploit is a piece of software, a chunk of data, or a sequence of commands that takes advantage of a vulnerability
Malware delivers a malicious "payload" that performs a harmful function

Defenses Against Attacks - Principles

- Layering

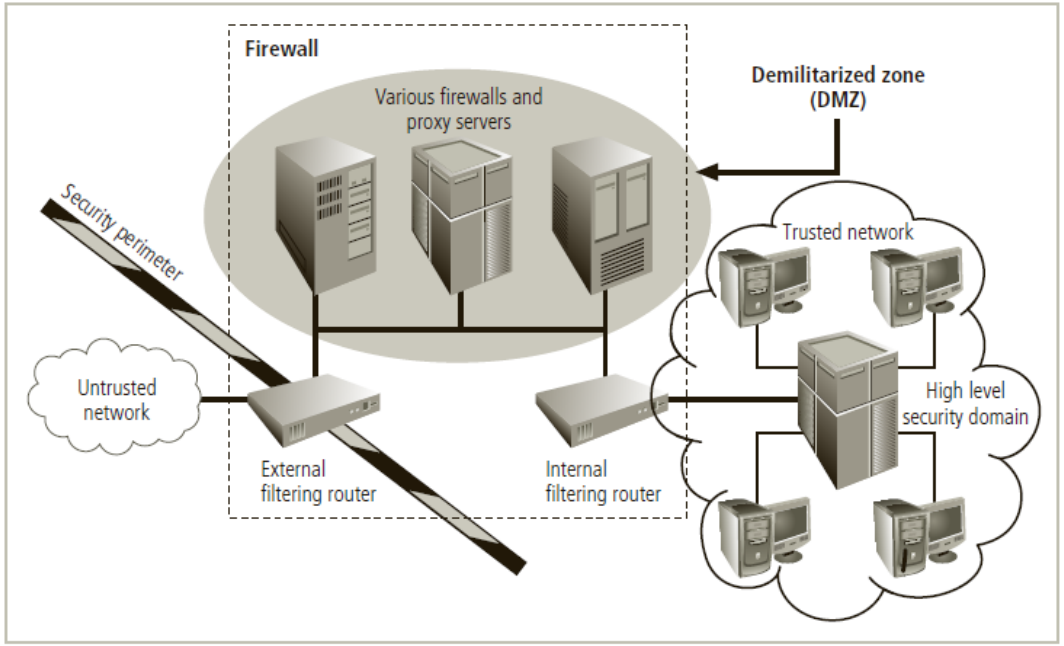
  If a layer is penetrated, several more layers must still be breached
- Limiting

  Only those who must use the data should have access to it; access should be limited to what those people need to perform
- Diversity

  Attackers cannot use the same technique to break multiple layers
- Obscurity

  Not revealing network topology, computer type, OS version
- Simplicity

  System should be simple for those on the inside to understand



Module 1 Activities

- Read Chapter 1 uploaded to Blackboard
- Homework Module 1